

CE 311S Exam 1 — Worked Examples & Attack Plans

Companion to the 2-page formula cheat sheet. Every problem type from the Spring 2026 review and the example midterm, fully worked. Practice writing these same steps on the real exam.

A. Descriptive stats — review Example 1 & sample-exam lakes

Review: $n = 20$ ages, $\Sigma x = 143$, $S_{xx} = 62.55$

Sorted: 4, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 10, 10

1. Mean $\hat{\mu} = 143/20 = 7.15$
2. n even $\rightarrow$ median = average of 10th and 11th = $(7+7)/2 = 7$
3. Lower half = first 10 values. $Q1$ = median of those = average of 5th and 6th of the lower half = $(6+6)/2 = 6$
4. $s^2 = S_{xx}/(n-1) = 62.55/19 = 3.292 \rightarrow s = 1.81$ (the review quotes $\text{Var} = 3.29$)
5. Histogram: $\sqrt{20} \approx 4.5 \rightarrow$ use 5 bins. Width = $(10-4)/5 = 1.2$. Bins $[4, 5.2], (5.2, 6.4], (6.4, 7.6], (7.6, 8.8], (8.8, 10]$ with counts 4, 5, 2, 3, 6.

Sample exam: mercury lakes 70, 85, 95, 80, 92, 97, 90, 100, 93, 95. $\Sigma x=897$, $\Sigma x^2=81,197$

1. Sorted: 70, 80, 85, 90, 92, 93, 95, 95, 97, 100
2. Mean = $897/10 = 89.7$
3. 10% trimmed mean: 10% of 10 = 1 from each end. Drop 70 and 100. Remaining sum = $897-170 = 727$. Mean = $727/8 = 90.875$
4. Median = average of 5th and 6th = $(92+93)/2 = 92.5$
5. $S_{xx} = 81197 - 897^2/10 = 81197 - 80460.9 = 736.1$. $s^2 = 736.1/9 = 81.789$. $s = 9.044$

On the exam: if they give Σx and Σx^2 , use them. Show S_{xx} then divide by $n-1$. For trimmed mean, state how many you dropped.

B. Counting — words, rebar, students

Order matters, no repeats

5-letter words from 26 letters, no repeats: $P(26,5) = 26 \times 25 \times 24 \times 23 \times 22 = 7,893,600$

Order matters, repeats allowed

4-letter words, repeats OK: $26^4 = 456,976$

Multivariate hypergeometric (sample exam rebar)

15 rebars: 5 of 1", 5 of 3/4", 5 of 1/2". Pull 6. P(two of each):

$$P = [C(5,2) C(5,2) C(5,2)] / C(15,6) = (10 \times 10 \times 10) / 5005 = 1000/5005 = 0.1998$$

Review Example 5 — 15 flu students, pick 3

4 seniors, 5 juniors, 6 freshmen. $C(15,3)=455$.

- Exactly 2 freshmen: $C(6,2)C(9,1)/455 = 135/455 = 0.2967$
- All 3 same category: $[C(4,3)+C(5,3)+C(6,3)]/455 = (4+10+20)/455 = 34/455 = 0.0747$
- One of each: $4 \times 5 \times 6 / 455 = 120/455 = 0.2637$

If the objects come in types and you sample without replacement, the fraction of combination products is almost always the answer. Write $C(\text{total}, n)$ in the denominator first so you don't forget it.

C. Bayes tree — review Example 2 & dam upgrade

Austine homes

Source A_i	$P(A_i)$	$P(B A_i)$	Joint $P(A_i \cap B)$	$P(A_i B)$
Solar A1	0.40	0.30	0.12	$0.12/0.455 = 0.2637$
Grid only A2	0.35	0.60	0.21	$0.21/0.455 = 0.4615$
Gas A3	0.25	0.50	0.125	$0.125/0.455 = 0.2747$

1. (a) $P(A2 \cap B) = 0.35 \times 0.60 = 0.21$
2. (b) $P(B) = 0.12+0.21+0.125 = 0.455$
3. (c) Three posterior probabilities above. They must sum to 1 (check: $0.2637+0.4615+0.2747=0.9999$).

Sample exam: $P(\text{upgraded} | \text{stable})$

$P(U)=0.20$, $P(S|U)=0.90$. From the filled table, 450 of 525 dams are stable, so $P(S)=450/525=6/7$.

$$P(U|S) = (0.90)(0.20) / (450/525) = 0.18 \times 525/450 = 0.21 = \mathbf{0.21}$$

Always say whether 0.21 is a probability (unitless). Box it. If they already gave you a table, use the table's $P(S)$ rather than making up $P(S | \text{not upgraded})$.

D. Independence of two RVs — review Example 3

Marginals of S and T both look like (0.25, 0.5, 0.25). That does **not** mean independence.

Test: $P(S=0)P(T=0) = 0.25 \times 0.5 = 0.125$, but the joint cell $P(S=0, T=0) = 0.25 \neq 0.125$.

Dependent.

You only need one failing cell. A faster visual: several joint cells are 0 while both corresponding marginals are positive — impossible for independent RVs.

E. Discrete RV + nonlinear $h(X)$ — review Example 4

25% of US bridges poor; sample $n=3$ at random, large population $\rightarrow$ **binomial** $X \sim \text{Bin}(3, 0.25)$.

x	C(3,x)	$p(x) = C(3,x)(0.25)^x(0.75)^{3-x}$
0	1	0.421875
1	3	0.421875
2	3	0.140625
3	1	0.015625

$$1. E[X] = np = 0.75. V(X) = np(1-p) = 0.5625. SD = 0.75.$$

$$2. E[X^2] = V + \mu^2 = 0.5625 + 0.5625 = 1.125$$

$$3. E[2X^2 + 3] = 2(1.125) + 3 = \mathbf{5.25}$$

$$4. E[X^4] = 0^4(0.421875) + 1^4(0.421875) + 2^4(0.140625) + 3^4(0.015625) = 0.421875 + 2.25 + 1.265625 = \mathbf{3.9375}$$

$$5. V(X^2) = E[X^4] - (E[X^2])^2 = 3.9375 - 1.265625 = 2.671875$$

$$6. V(2X^2 + 3) = 4 V(X^2) = \mathbf{10.6875}$$

You cannot use $V(a+bX) = b^2 V(X)$ directly on $2X^2 + 3$ because it is not linear in X . Rewrite as $a+bY$ with $Y=X^2$, then $V=b^2 V(Y)$.

F. E and SD of a function of a listed pmf (sample exam high-rises)

HB takes 3, 5, 7 with probabilities 0.35, 0.45, 0.20. Cost $C = h(\text{HB})$ (whatever quadratic/linear they print).

1. Make a three-row table: hb, p, $c=h(\text{hb})$, c^2 .

2. $E[C] = \sum c p$. Sample answer was **\$806.17 million**.

3. $E[C^2] = \sum c^2 p$, then $SD = \sqrt{E[C^2] - (E[C])^2}$. Sample answer **\$167.6 million**.

4. Because h is almost certainly not linear, do **not** compute $h(E[\text{HB}])$. $E[\text{HB}] = 4.7$, and $h(4.7)$ is the wrong expected cost.

G. Contingency table algebra (sample exam dams)

Urban given: Old-S 100, Old-U 30, New-S 200, New-U 20. Urban total = 350.

- $1/3$ of all dams rural $\Rightarrow$ urban is $2/3 \Rightarrow$ Total = $350 \times 3/2 = 525$. Rural total = 175.
- 50 older dams are unstable. Urban old-unstable = 30 $\Rightarrow$ rural old-unstable = 20.
- $1/6$ of older rural dams are unstable $\Rightarrow$ rural old total = $20 \times 6 = 120 \Rightarrow$ rural old-stable = 100.
- $1/3$ of all stable dams are rural. Urban stable = 300 $\Rightarrow$ all stable = 450 $\Rightarrow$ rural stable = 150 $\Rightarrow$ rural new-stable = 50.
- Rural new total = $175 - 120 = 55 \Rightarrow$ rural new-unstable = 5.

Rural row: **100, 20, 50, 5**. $P(\text{unstable} | \text{rural}) = 25/175 = \mathbf{0.1429}$.

Exactly 5 of 8 randomly selected older dams stable:

- Older: 200 stable + 50 unstable = 250.
- Exact hypergeometric: $C(200,5)C(50,3)/C(250,8) \approx \mathbf{0.1471}$
- Binomial approx $p=0.8$: $C(8,5)(0.8)^5(0.2)^3 = 56 \times 0.32768 \times 0.008 = \mathbf{0.1468}$

On Kockelman's exam, either answer is typically accepted if you name the distribution. The binomial is faster on a calculator. Mention " $n \ll N$ so $p \approx \text{constant}$ " if you approximate.

H. Estimating n and p (sample exam EPA lakes)

Treat yearly mercury-lake count as $X \sim \text{Bin}(n,p)$. Method of moments:

$$\begin{aligned}\bar{x} &= 89.7 \approx np \\ s^2 &= 81.789 \approx np(1-p) \\ 1-p &\approx s^2/\bar{x} = 81.789/89.7 = 0.91181 \rightarrow \hat{p} = 0.08819 \\ \hat{n} &= 89.7 / 0.08819 \approx 1017\end{aligned}$$

Extra credit — why binomial may be wrong: contamination is spatially clustered (trials not independent); the set of U.S. lakes is finite (should be hypergeometric); p can trend over 10 years; EPA may not use simple random sampling; n may not actually be constant.

I. What to write in the first 60 seconds

1. cdf = cumulative distribution function
2. Bayes: $P(A_i | B) = P(B | A_i)P(A_i) / \sum_j P(B | A_j)P(A_j)$
3. $E[h(X)] = h(E[X])$ iff $h(x) = a + bx$ (linear)
4. Dump the formula block from the cheat sheet onto scrap paper in case you cannot bring this sheet in.

Kockelman's cover note: show all calculations, state assumptions, include units, and box numerical answers. A right number with no box / no units / no distribution named will leak points.

J. Night-before 10-question drill (cover the answers)

#	Prompt	Answer
1	What does cdf stand for?	Cumulative distribution function, $F(x) = P(X \leq x)$
2	Write Bayes' complete theorem.	$P(A_i B) = P(B A_i)P(A_i) / \sum_j P(B A_j)P(A_j)$
3	When is $E[h(X)] = h(E[X])$?	Only if h is linear: $h(x) = a + bx$
4	5-letter words, no repeated letters, 26-letter alphabet?	$P(26, 5) = 26 \times 25 \times 24 \times 23 \times 22 = 7,893,600$
5	How do you prove two RVs are independent?	$p(s, t) = p(s)p(t)$ for every cell. One miss $\rightarrow$ dependent
6	Pick 6 rebars from 5+5+5 types, want 2 of each?	$C(5, 2)^3 / C(15, 6) = 1000/5005 = 0.1998$
7	$X \sim \text{Bin}(n, p)$, data give $\bar{x}$ and s^2 . Find p and n.	$\hat{p} = 1 - s^2/\bar{x}$ $\hat{n} = \bar{x}/\hat{p}$
8	Finite population, without replacement?	Hypergeometric (or multivariate hypergeometric)
9	$V(a + bX)$ and $SD(a + bX)$?	$b^2V(X)$ and $ b \sigma$. Does not apply to $2X^2 + 3$
10	Kockelman presentation rules?	Show work, state assumptions, UNITS, BOX the number

Print the 2-page formula sheet double-sided. Use this packet to drill, then put both PDFs in your CE 311S course folder.